

DR. DEBJIT DE

Assistant Professor, Department of Bioinformatics, School of Bioengineering and Biosciences, Lovely Professional University

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Kolkata

https://debjitde01.github.io/portfolio-github/

Summary

Motivated bioinformaticist with wet-lab and computational genomics competence. Highly skilled in DNA extraction, PCR, and microbial culturing, as well as high-throughput microbiome and functional omics data analysis using QIIME2, DADA2, MetaPhlAn, and HUMAnN2. Loves unraveling hostmicrobe relationships and studying microbial diversity using molecular and in silico methods to advance translational research.

Experience

Swami Vivekananda University — Barrackpore, West Bengal, India

Assistant Professor — September, 2023 - July, 2026

- Leading research initiatives in NGS-based microbiome data analysis, integrating bioinformatics pipelines and statistical meta-analyses.
- Supervising student thesis work and mentoring research scholars through proposal development, data interpretation, and publication processes.
- Teaching undergraduate and postgraduate courses in biotechnology, with a proactive role in curriculum design, assessment strategy, and instructional innovation.
- Worked as an exam coordinator for the Department of Biotechnology, where I handled all exam-related tasks like rules & regulation making, BOS for the exam, exam routine setup, question paper setup and finalisation with exam controler, tabulation of marksheet etc.
- Coordinating academic logistics, including examination planning, faculty collaboration, and administrative oversight to ensure smooth academic operations.

National Institute of Biomedical Genomics — Kalyani, West Bengal, India

Senior Research Data Analyst — April, 2022 - August, 2023

- Conducted high-throughput bioinformatics and statistical analyses on 16S rRNA and whole metagenome sequencing datasets to uncover microbiome patterns and biological insights.
- Ensured robust data integrity through systematic cleaning, validation, and preprocessing of complex genomic datasets.
- Translated data into informative visual narratives using R, Python, and other tools to support hypothesis generation and scientific communication.
- Authored and contributed to scientific manuscripts, presentations, and internal reports to disseminate findings to both technical and nontechnical audiences.
- Oversaw the curation and secure management of large-scale genomic data, maintaining compliance with institutional and ethical data governance standards.

Awards

JU-RUSA 2.0
February 2019 - March 2020
MHRD, Govt. of India
JU-RUSA 2.0 Doctoral Scholarship Scheme

Certifications

Computational Genomics
May 2025
NPTEL - AICTE
Obtained 96% - Topper 1%

BioInformatics: Algorithms and Applications
May 2025
NPTEL - AICTE
Obtained 71%

Next Generation Sequencing Technologies : Data Analysis and Applications
October 2025
NPTEL - AICTE
Obtained 57%

Research Methodology
March 2026
NPTEL - AICTE
Obtained 62%

Languages

ENGLISH
Speak, Read, Write
●●●●●

HINDI
Only Speak
●●●○○

BENGALI
Speak. Read, Write
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Education

Jadavpur University Ph.D.
Computational Biology, Applied Microbiology 2015 – 2023
Research project on "*Characterization of gut microbiota of the Indian Type 2 Diabetic Patients: Sustainability of Human Health*" under the supervision of **Dr. Paltu Kumar Dhal**, Department of Life Science and Biotechnology, Jadavpur University, Kolkata – 700032, INDIA.

Vellore Institute of Technology (V.I.T.) University M.Sc. • 7.5
Biomedical Genetics 2012 – 2014

Institute of Genetic Engineering College, M.A.K.A.U.T. B.Sc.(H) • 7.49
Genetics 2009 – 2012

Projects

Research 2014 – 2015

- Research project on "An In-Silico Approach to Identify Secondary tRNA Structure" under the supervision of Prof. Subhasis Mukhopadhyay, Department of Bio-Physics, Molecular Biology and Bioinformatics, University of Calcutta, Kolkata – 700 009, INDIA, 2014.
- Research project on "Exploring the Correlation Between Habitat Growth Temperature & Genomic Conformation in Bacterial Population: An insilico Approach" under the supervision of Prof. Subhasis Mukhopadhyay, Department of Bio-Physics, Molecular Biology and Bioinformatics, University of Calcutta, Kolkata – 700 009, INDIA, 2014.

Project 2012 – 2014

- Research project on "In-silico Analysis of Huntington's Disease Gene" under the supervision of Prof. Mahalingam K., Associate Professor, School of Bio Science & Technology, Vellore Institute of Technology (V.I.T.) University, Vellore – 632014, Tamil Nadu, INDIA, 2013.
- Research project on "Genotoxic Evaluation of Commonly Used Drugs in Model Systems (Drosophila melanogaster)" under the supervision of Prof. Sonali Sengupta, Assistant Professor, School of Bio Science & Technology, Vellore Institute of Technology (V.I.T.) University, Vellore – 632014, Tamil Nadu, INDIA, 2013.
- Research project on "Biosynthesis of Nanoparticles (Silver) From Juglans regia & Tribullus terrestris And Genotoxicity Test" under the supervision of Prof. Gopiesh Khanna V, Assistant Professor, School of Bio Science & Technology, Vellore Institute of Technology (V.I.T.) University, Vellore – 632014, Tamil Nadu, INDIA, 2012.

Publications

Research Articles

1. Basak T, Banik RD, **De D**, Pal P, Pal P. Tissue-Specific Arsenic Bioaccumulation in Labeo Rohita and Catla Catla Across Contaminated Districts of West Bengal, India. Bull Environ Contam Toxicol. 2026 Mar 7;116(3):50. doi: 10.1007/s00128-026-04210-5. PMID: 41793445. Impact Factor - 2.2
2. Dey, N., & Bandyopadhyay, B., **De, D.** (2025). Single-Run method for Antibiotics in Honey through UPLC-MS/MS. *Indian Journal of Pharmaceutical and Biological Research*, *13*(03), 1-4. https://doi.org/10.30750/ijpbr.13.3.01.

Skills

Computational Biology
QIIME, Trimmomatics, PEAR, SINA, MOTHUR, SWARM, QIIME-2, DADA2, HUMANN-2, MetaPhlan, SORTMERNA, DESeq2

Machine learning model build and prediction
Supervised and Unsupervised Model, CrossValidation, Bootstrap

Computer
Linux, Microsoft Office

Programming Language
C-programming, Perl, R Programming, Python - Intermediate, HTML

Research Interest

- Prokaryotic genome sequence analysis
- Microbial metagenomics and host pathogen interactions
- Meta-analysis and machine learning study of human disease
- Human disease multi-omics analysis
- AMR study
- Metagenomics (viromics) phage bacteria interaction

Teaching Interest

- Introduction to Bioinformatics (Theory and Laboratory)
- Molecular Evolution
- Genomics and Proteomics
- Molecular Modelling and Drug Design (Theory)
- Perl and BioPerl
- R Programming for Bioinformatics

3. Kunda, P., Mondal, S., **De, D.**, Dhal, P. K., & Mukherjee, A. (2024). Alteration of Rice Root Endophytic Bacterial Community Composition by Meloidogyne graminicola and Identification of Potential Biocontrol Agent. Annals of Microbiology, 74(1), 42. Impact Factor – 3.0

4. Talukdar, D., Sarkar, M., Ahrodia, T., Kumar, S., **De, D.**, Nath, S., ... & Das, B. (2024). Previse preterm birth in early pregnancy through vaginal microbiome signatures using metagenomics and dipstick assays. iScience. Impact Factor – 4.6

5. Nath, S., Sarkar, M., Maddheshiya, A., **De, D.**, Paul, S., Dey, S., Pal, K., Roy, S.K., Ghosh, A., Sengupta, S. and Paine, S.K., 2023. Upper respiratory tract microbiome profiles in SARS-CoV-2 Delta and Omicron infected patients exhibit variant specific patterns and robust prediction of disease groups. Microbiology Spectrum, 11(6), pp.e02368-23. Impact Factor – 9.0

6. **De, D.**, Nayak, T., Chowdhury, S., & Dhal, P. K. (2022). Insights of host physiological parameters and gut-microbiome of Indian Type 2 Diabetic patients visualized via metagenomics and machine learning approaches. Frontiers in Microbiology, 0, 2479. doi.org/10.3389/FMICB.2022.914124. Impact Factor – 4.0

7. Nayak, T., **De, D.**, Dhal, P. K. (2021). The differences in SARS-CoV and SARS-CoV-2 specific co-expression network mediated biological process in human gut enterocytes. Infection, Genetics and Evolution, doi.org/10.1016/j.meegid.2021.104892. Impact Factor – 2.6

8. Nayak, T., **De, D.**, Karmakar, P., Deb, A., Dhal, P. K. (2021). Microbial communities of the drinking water with gradient radon concentration are primarily contributed by radon and heavy metal content. Frontiers in Environmental Science, 9, doi: 10.3389/fenvs.2021.576400. Impact Factor – 3.3

9. Nayak, T., **De, D.**, Barman, C., Karmakar, P., Deb, A., Dhal, P. K. (2019). Characterization of indigenous bacteria from radon-rich groundwater and their tolerance to physicochemical stress. International Journal of Environmental Science and Technology, 17, 1627–1636, https://doi.org/1 0.1007/s13762-019-02445-w. Impact Factor – 3.0

Book Chapters

1. Banerjee, S., Adhikary, P., De, D., Kalam, A., Chowdhury, S., & Mal, C. (2026). Role of the gut microbiome in the pathogenesis of inflammatory bowel disease. In The gut microbiome in precision medicine: From research to clinical practice (pp. 341–353). Academic Press. https://doi.org/10.1016/C2024-0-03206-7

2. De, D., Nayak, T., Das, G., & Dhal, P. K. (2023). Metagenomics and bioinformatics in microbial ecology: Current status and beyond. Applications of Metagenomics, 359-385. https://doi.org/10.1016/B978-0 323-98394-5.00009-2, ELSEVIER, ISBM no. – 9780323983945

3. Sutradhar, B.D., Ghosh, S., De, D. (2024) Chapter 1 – GBA Gene Expression Study Of Parkinson's Disease. Research Renaissance: Integration of Interdisciplinary Terrains (pp. 1-09). ISBN No. – 979 889588010-4

Patent

1. Published patent on Biopesticide Magic Spray, Application ID & Date 202531022109, 12/03/2025; Published Date - 21/03/2025

References

Dr. Paltu Kumar Dhal
Assistant Professor
Department of Life Science and Biotechnology

Jadavpur University

E-mail: paltuk.dhal@jadavpuruniversity.in

Contact Number: +91-9051080809

Dr. Chittabrata Mal
Associate Professor
Department of Bioinformatics

Maulana Abul Kalam Azad University of Technology, West Bengal

E-mail: chittabrata@gmail.com

Contact Number: 09163482045

Prof. Subhasis Mukhopadhyay
Professor (Retired)
Department of Biophysics, Molecular

Biology and Bioinformatics

University of Calcutta

E-mail: sm.bmbg@gmail.com

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Volunteer

Research Scholar Guidance

Guided **Mr. Navaneet Dey**, Ph.D. Research Scholar, as the **Research Supervisor** for the successful submission of his doctoral thesis entitled **"Physiological Assessment of Different Types of Honey and Development of a New Protocol for Multi Residual Menaces."** The research was carried out under the supervision of **Dr. Debjit De**, Assistant Professor, Department of Biotechnology, School of Life Sciences, Swami Vivekananda University, with **Prof. (Dr.) Bidyut Bandyopadhyay**, Professor, Department of Biotechnology, Oriental Institute of Science and Technology, Vidyasagar University, serving as Co-Supervisor.

Organiser

1. Trained 2 Students for internship (2 Months) on bioinformatics genome sequence analysis pipeline - one student came from M.Sc Biotechnology, Presidency University, West Bengal and another student came from M.Sc. Biotechnology, Chandigarh University, Chandigarh.
2. Trained 1 student of B.Sc. Biotechnology, Kanchrapara College, for internship on Basics of Bioinformatics - 1 Week
3. Trained 2 Students for internship (2 Months) on bioinformatics genome sequence analysis pipeline - one student came from M.Sc Biochemistry, University of Calcutta, West Bengal and another student came from B.Sc. Bioinformatics, Asian University For Women, Bangladesh.
4. Organizer of 6-Days Hands-on Training & Workshop on ***NextGen Bioresearch: Tools & Techniques in Resarch Methodology & Computational Biology***, Swami Vivekananda University, Barrackpore, India, 19th August– 22th August, 2025.
5. Organizer of 6-Days Hands-on Training & Workshop on ***Advanced Biological Instruments: Enhancing Research and Practical Skills***, Swami Vivekananda University, Barrackpore, India, 21st July – 26th July, 2025.
6. Organizer of Ten Days Hands-on Training on ***Drug Discovery: From Bench to Bedside***, Swami Vivekananda University, Barrackpore, India, 25th February – 4th March, 2025.
7. Organizer of One Day Hands-on Training on ***Next Generation Sequence (NGS) Data Handling***, Swami Vivekananda University, Barrackpore, India, 13th June, 2024.
8. Organizer of hands-on training on ***Next Generation Sequence (NGS) Data Handling, Metagenomics Study: Approaches (16S Amplicon & Whole Genome) and Analysis Using R – Programming Language***, Swami Vivekananda University, Barrackpore, India, January, 2024 – March, 2024.
9. Organizer of workshop on ***Human Microbiome Data Analysis***, National Institute of Biomedical Genomics (NIBMG), Kalyani, India, December 2022

Faculty Development

1. Attend Faculty Development Programme organized by School of Life Sciences, Swami Vivekananda University – 7th and 19th of July 2025.
2. Attend Faculty Development Programme organized by Swami Vivekananda University and Adamas University – 6th, 9th, 10th, 13th and 15th of May, 2024.
3. Attend Faculty Development Programme organized by Swami Vivekananda University – 10th and 11th of February 2024.
4. Attend Faculty Development Programme organized by Swami Vivekananda University – 23rd and 24th of September 2023.

Personal Information

- **Father's Name: Dilip Kumar De**
- **Mother's Name: Bithika De**
- **Brother's Name: Debaditya De**
- **Date of Birth: 05.02.1992**
- **Gender: Male**
- **Nationality: Indian**
- **Religion: Hindu**
- **Category: General**
- **Marital Status: Unmarried**
- **Address: B-86, New Garia Co. – Op. Housing Development Society, Kolkata – 700094, West Bengal, India**